

High level programming language for quantum computing

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Obiettivo del lavoro

Reasoning su quantum computer

Base di conoscenza classica + query

Embedding su quantum computer

Esecuzione e recupero risultati

Perché il quantum computing

Grazie a

- Superposition
- Entanglement

Violiamo

Strong Church-Turing
Thesis

$$\mathcal{H}(t) |\psi(t)\rangle = i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle$$

$$\langle A_0 \otimes B_0 \rangle + \langle A_0 \otimes B_1 \rangle + \langle A_1 \otimes B_0 \rangle - \langle A_1 \otimes B_1 \rangle = 2\sqrt{2}$$

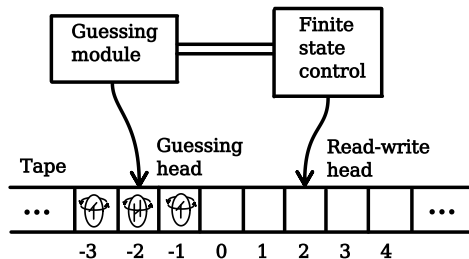


Figure: Probabilistic Turing Machine

Due paradigmi di programmazione

Quantum Gate

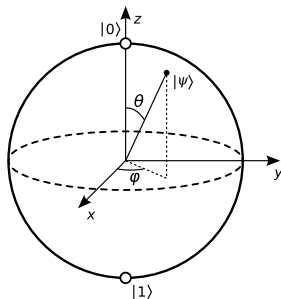


Figure: A single *qubit* gate can be seen as a rotation on the Bloch Sphere

Quantum Annealing

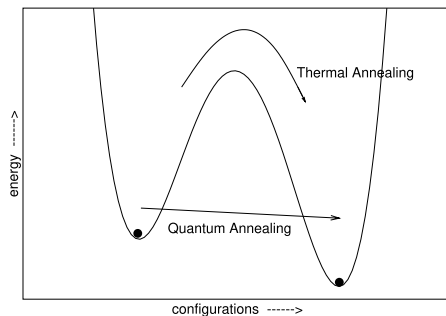


Figure: Tunneling effect (K. Chakrabarti, A. Das)

Quantum Gate

- Formalismo molto studiato
- Gate Reversibili
- Set di porte universali
- Algoritmi di Shore, ecc.

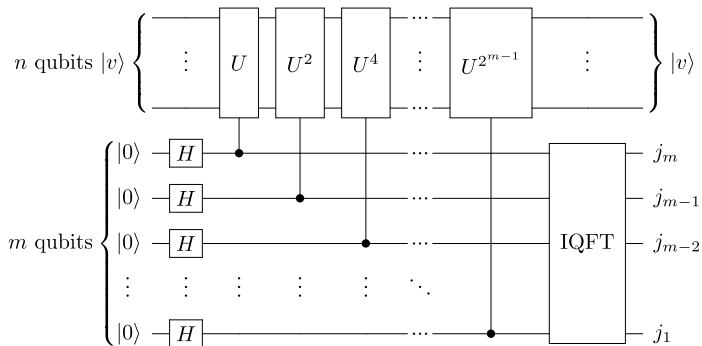
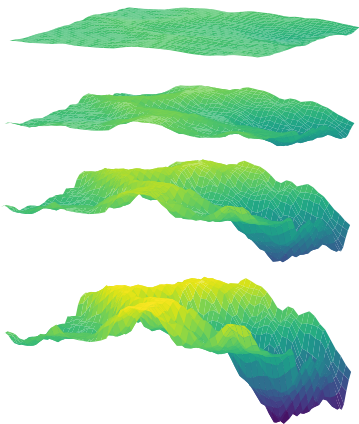


Figure: Step for Shor algorithm (Thomas G. Wong)

Quantum Annealer



- Problemi in forma QUBO
- Ispirato a Simulated Annealing
- Effetto tunneling
- Adiabatic Quantum Computing (AQC)
- $\mathcal{H}(t) = s(t)\mathcal{H}_i + (1 - s(t))\mathcal{H}_f$

Figure: Complete sovrapposizione $\rightarrow$ ground state $\mathcal{H}_f$

QA-Prolog

Il progetto

- Sviluppato da Scott Pakin
- Python + GO
- Proof of concept
- Trasformazioni successive
- Prolog + Query $\rightarrow \mathcal{H}_f$
- Interagisce direttamente con il solver
- Raccoglie e organizza i risultati

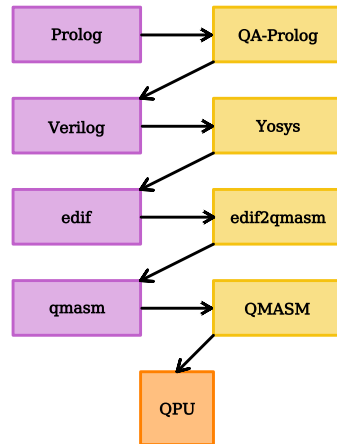


Figure: Transformation pipeline

QMASM

$\mathcal{H}_f$ simbolica $\rightarrow \mathcal{H}_f$ fisica

Cos'è

- Quantum macro assembler
- Si interfaccia con Ocean

Cosa permette di fare

- Riferimento simbolico a *qubit*
- *Qubit* “pinnati” a TRUE o FALSE
- Incapsulare pattern in macro
- Creazione di librerie di macro
- Organizzazione output

```
# Y = A OR B
!begin_macro OR
$A 0.5
$B 0.5
$Y -1
```

```
$A $B 0.5
$A $Y -1
$B $Y -1
!end_macro OR
```

Figure: or gate

Creiamo il file `gates.qasm`

Yosys - edif2qmasm

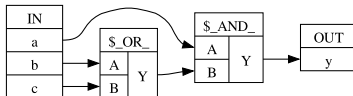
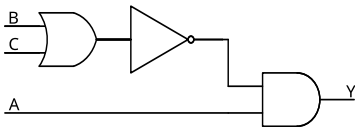
Verilog → Circuito digitale → $\mathcal{H}$ simbolica

Yosys

- Framework per la sintesi del Verilog
- Output: RTL Netlist in formato EDIF

edif2qmasm

- Converte dal formato EDIF a QMASM
- Attinge a una libreria di gate



```
!begin_macro sat
  !use_macro AND $id00006
  !use_macro NOT $id00004
  !use_macro OR $id00005
  $id00004.A = $id00005.Y
  $id00005.A = b
  $id00005.B = c
  $id00006.A = a
  $id00006.B = $id00004.Y
  $id00006.Y = y
!end_macro sat
```

QA-Prolog

- Traduzione da Prolog a Verilog
- Wrapper per tutta la Pipeline
- Risultati formato Human Readable
- Dimensionamento automatico variabili

```
sat(A, B, C, Y) :-  
    or(B, C, X),  
    not(X, Z),  
    and(A, Z, Y).
```

Figure: Prolog Code

```
module sat (A, B, C, Y);  
    input A, B, C;  
    output Y;  
  
    assign Y = A & ~(B | C);  
endmodule
```

Figure: Verilog Code

Interrogazione

```
QA-Prolog --qasm-args="--solver=tabu --postproc=opt"  
--query='sat(A, B, C, true).' circ_sat.pl
```

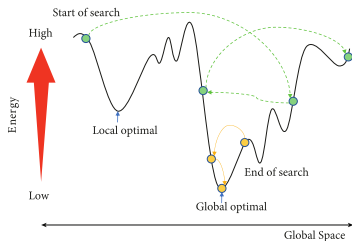
```
A = true  
B = false  
C = false
```

Figure: Query Result

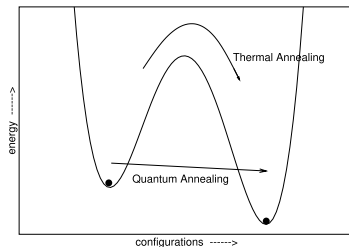
Non un solo paradigma

Problema QUBO/Ising

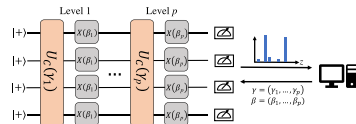
$$\mathcal{H}_f$$



(a) Simulated annealing (Li Yang)



(b) Quantum annealing (K. Chakrabarti, A. Das)



(c) QAOA (Z. Zhou, Y. Du, X. Tian, D. Tao)

Figure: Single representation 3 different approaches

QAOA

Quantum approximate optimization algorithm

- 1 Definire matrici: $\mathcal{H}_c$ e $\mathcal{H}_m$
- 2 Definire oracoli parametrici: $\mathcal{U}_c(\gamma) = e^{-i\gamma\mathcal{H}_c}$ e $\mathcal{U}_m(\beta) = e^{-i\beta\mathcal{H}_m}$
- 3 Applicazione ripetuta di $\mathcal{U}_c(\gamma)$ e $\mathcal{U}_m(\gamma)$
- 4 Ottimizzazione (classica) dei parametri γ_i e β_i

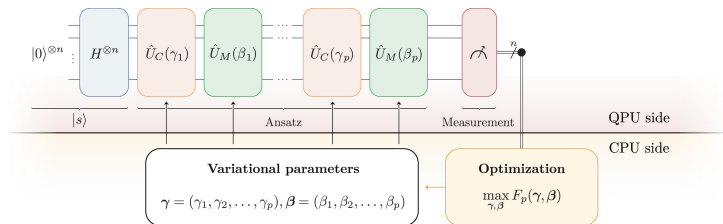
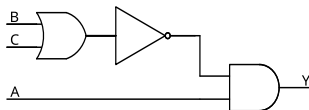


Figure: QAOA sketch (BLEKOS, Kostas, et al.)

Esempio



```

!begin_macro sat
!use_macro AND $id000006
!use_macro NOT $id000004
!use_macro OR $id000005
$id000004.A = $id000005.Y
$id000005.A = b
$id000005.B = c
$id000006.A = a
$id000006.B = $id000004.Y
$id000006.Y = y
!end_macro sat

```

Figure: CircSat

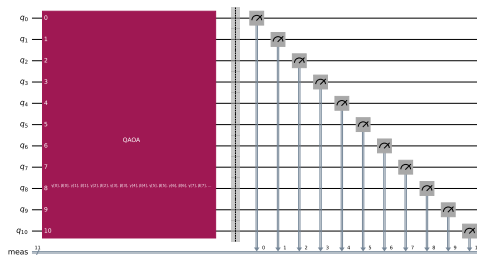
$$\begin{pmatrix}
 0.5 & 0.5 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 \\
 0 & 0.5 & -1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 1 & 0 & 0 & -2 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & -0.5 & 0.5 & -1 & -2 & 0 & 0 \\
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 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{pmatrix}$$


Figure: $\mathcal{H}_f$ and QAOA Quantum Circuit

Esempio

Risultato 2

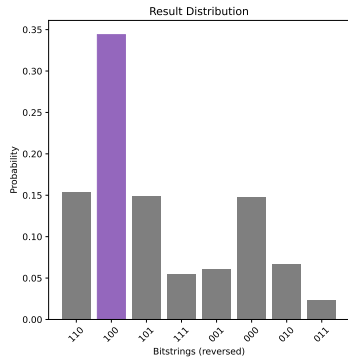


Figure: 3 “interesting” bit

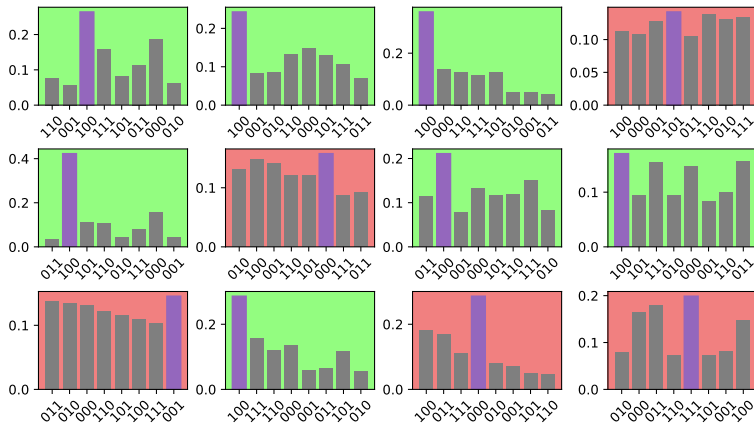


Figure: Multiple experiment runs

Ontologia

I Simpson

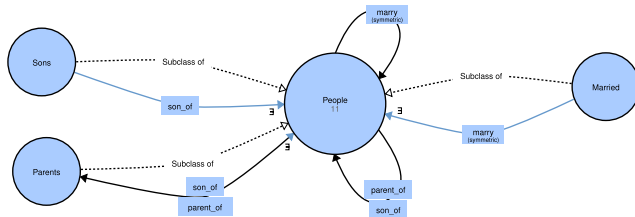


Figure: Ontology Structure

```
<owl:NamedIndividual rdf:about="http://www.people#marge">
  <rdf:type rdf:resource="http://www.people#People"/>
  <www:marry rdf:resource="http://www.people#homer"/>
  <www:parent_of rdf:resource="http://www.people#bart"/>
</owl:NamedIndividual>
```

Figure: Marge Simpson entry

Da OWL-rdf a Prolog

```
person(marge).  
person(bart).  
person(jackie).  
person(selma).  
person(ling).
```

```
parent_of(marge, bart).  
parent_of(jackie, marge).  
parent_of(jackie, selma).  
parent_of(selma, ling).
```

```
son_of(P, Q) :- parent_of(Q, P).
```

```
grand_parent_of(P, Q) :-  
    son_of(Q, X),  
    son_of(X, P).
```

Figure: Prolog Ontology

Interrogazione

```
QA-Prolog --verbose --qasm-args="--solver=sim_anneal  
--postproc=opt" --query='grand_parent_of(P1, P2).' family.pl
```

```
P1 = jackie  
P2 = bart
```

```
P1 = jackie  
P2 = selma
```

Figure: Query Result 1

```
P1 = jackie  
P2 = bart
```

Figure: Query Result 2

Lavoro Futuro

Monte

- Compilatore OWL/RDF → Prolog
- Sperimentare altri linguaggi

Valle

- Aggiungere simulated annealing
- Interfacciare QMASM con quantum gate

- Mantenere aggiornati i componenti
- Testare su hardware quantistico
- Confronto ACQ e QAOA

Grazie per l'attenzione

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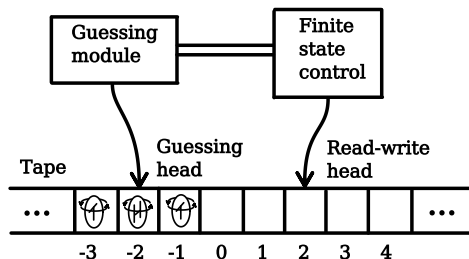


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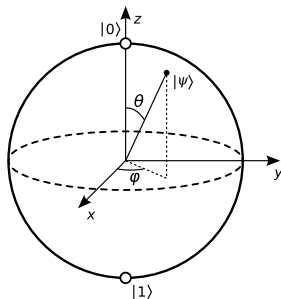


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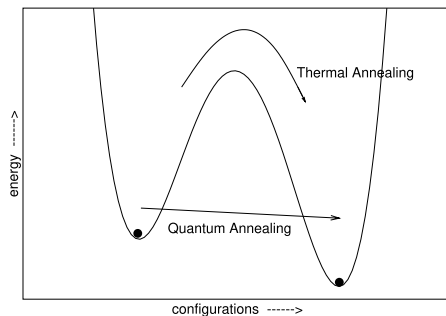


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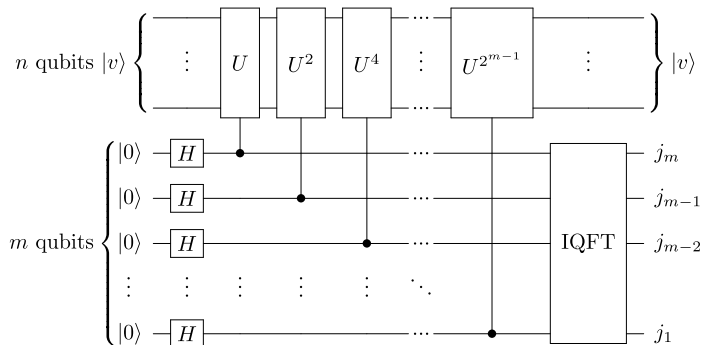
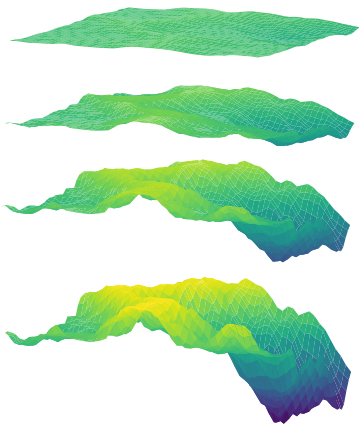


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Figure: Complete sovrapposizione $\rightarrow$ ground state $\mathcal{H}_f$

QMASM

Esempio: Macro

```
# Y = A OR B
!begin_macro OR
  $A 0.5
  $B 0.5
  $Y -1

  $A $B 0.5
  $A $Y -1
  $B $Y -1
!end_macro OR
```

Figure: or gate

```
# Y = NOT A
!begin_macro NOT
  $A $Y 1.0
!end_macro NOT
```

Figure: not gate

```
# Y = A AND B
!begin_macro AND
  $A -0.5
  $B -0.5
  $Y 1

  $A $B 0.5
  $A $Y -1
  $B $Y -1
!end_macro AND
```

Figure: and gate

possiamo racchiudere queste macro nel file `gates.qasm`

QMASM

Esempio: $y = x_1 \wedge \neg(x_2 \vee x_3)$

```
!include <gates>

!use_macro OR x2_or_x3
  x2_or_x3.$A = x2
  x2_or_x3.$B = x3
  x2_or_x3.$Y = $x4

!use_macro NOT not_x4
  not_x4.$A = $x4
  not_x4.$Y = $x5

!use_macro AND x1_and_x5
  x1_and_x5.$A = x1
  x1_and_x5.$B = $x5
  x1_and_x5.$Y = y
```

Figure: CircSat problem

“Pinniamo” il valore di y per ottenere l’assegnamento delle x_i che verificano la formula logica:

```
qasm --run --pin="y := true" circsat.qasm
```

Solution #1 (energy = -20.0000, tally = 647):

Variable	Value
-----	-----
x1	True
x2	False
x3	False
y	True

Figure: CircSat solution

Yosys - edif2qasm

Esempio: moltiplicazione tra interi

```
module mult (multiplicand, multiplier, product);  
  input [1:0] multiplicand;  
  input [1:0] multiplier;  
  output [2:0] product;  
  
  assign product = multiplicand * multiplier;  
endmodule
```

Figure: Factorization problem

Traduzione in EDIF

```
yosys myfile.v synth.ys -b edif -o myfile.edif
```

Traduzione in QMASM

```
edif2qasm -o="myfile.qasm" myfile.edif
```

Esecuzione

```
qasm --run --pin="mult.product[2:0] := 110"  
--solver="sim_anneal" mult.qasm
```

Solution #1 (energy = -57.5000, tally = 68):

Variable	Value
mult.multiplicand[0]	False
mult.multiplicand[1]	True
mult.multiplier[0]	True
mult.multiplier[1]	True
mult.product[0]	False
mult.product[1]	True
mult.product[2]	True

Figure: Factorization Solution

QA-Prolog

Esempio

Base di conoscenza

Il nemico del mio nemico è mio amico

```
hates(alice, bob).  
hates(bob, charlie).  
  
enemies(P, Q) :- hates(P, Q).  
enemies(P, Q) :- hates(Q, P).  
  
friends(A, B) :-  
    enemies(A, X),  
    enemies(X, B),  
    A \= B.
```

Figure: Enemy of my Enemy

Esecuzione

```
QA-Prolog --qasm-args="--postproc=opt"  
--query='friends(P1, P2).' friends.pl
```

```
P1 = alice  
P2 = charlie
```

```
P1 = charlie  
P2 = alice
```

Figure: Query Solution

Update al progetto

Integrazione con Ocean

- Aggiornamento librerie obsolete
- Rimozione metodi deprecati
- Sostituzione funzioni “spostate”
- Correzione parametri funzioni

Altro

- Sostituzione funzioni rinominate
- Parametri command line

```
#from dwave.cloud import Client, hybrid, qpu, sw
from dwave.cloud import Client
import hybrid
```

```
#from greedy import SteepestDescentSolver
from dwave.samplers import SteepestDescentSolver
```

```
#from tabu import TabuSampler
from dwave.samplers import TabuSampler
```

```
#from scipy.stats import median_absolute_deviation
from scipy.stats import median_abs_deviation
```

Figure: Update librerie

```
#elif solver == "neal":
elif solver == "sim_anneal":
```

Figure: Correzione parametri command line

Update al progetto

Incompatibilità tra output edif2qasm e input QMASM

```
// Define hates(atom, atom).  
module \hates/2 (A, B, Valid);  
...  
endmodule
```

Figure: Verilog

```
# hates/2  
!begin_macro id00011  
...  
!end_macro id00011  
  
# enemies/2  
!begin_macro id00010  
!use_macro OR $id00031  
!use_macro hates/2  
...
```

Figure: qasm

```
with open(file, 'r') as input:  
    first = input.readline()  
    second = input.readline()  
    while(second_row != ""):  
        if first.startswith("#") and second.startswith("!begin_macro"):  
            self.name[first[2:-1]] = second[len("!begin_macro "):-1]  
            first_row = second_row  
            second_row = input.readline()  
  
with open(file, 'r') as input:  
    doc = input.read()  
    lines = doc.splitlines()  
    for line in lines:  
        for word in line.split():  
            if word in self.name.keys():  
                line = line.replace(word, self.name[word])  
            self.new_lines.append(line)
```

Figure: Preprocessing

QAOA

Quantum approximate optimization algorithm

- 1 Definire matrici: $\mathcal{H}_c$ e $\mathcal{H}_m$
- 2 Definire oracoli parametrici: $\mathcal{U}_c(\gamma) = e^{-i\gamma\mathcal{H}_c}$ e $\mathcal{U}_m(\beta) = e^{-i\beta\mathcal{H}_m}$
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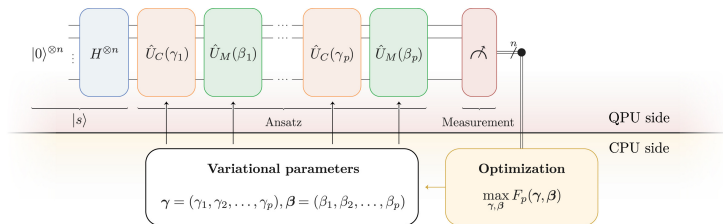


Figure: QAOA sketch (BLEKOS, Kostas, et al.)

QUBO $\rightarrow$ ISING $\rightarrow$ Operatori di Pauli

```
qubo = qubovert.utils.matrix_to_qubo(qubo_mat)
ising = qubovert.utils.qubo_to_quso(qubo)
qubo_dict = dict(qubo)
ising_dict = dict(ising)
ising_dict.pop(())
ising_mat = qubovert.utils.qubo_to_matrix(ising_dict)
```

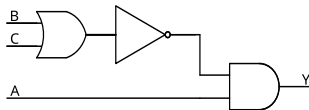
Figure: QUBO $\rightarrow$ ISING (libreria qubovert)

```
def build_paulis(matrix):
    pauli_list = []
    for i in range(len(matrix)):
        pauli_list.append(("Z", [i], matrix[i][i]))
        for j in range(i+1, len(matrix)):
            pauli_list.append(("ZZ", [i, j], matrix[i][j]))
    return pauli_list

sat_paulis = build_paulis(ising_mat)
cost_hamiltonian = SparsePauliOp.from_sparse_list(sat_paulis, n_qubits)
```

Figure: ISING $\rightarrow$ Operatori di Pauli

Esempio



```

!begin_macro sat
  !use_macro AND $id00006
  !use_macro NOT $id00004
  !use_macro OR $id00005
  $id00004.A = $id00005.Y
  $id00005.A = b
  $id00005.B = c
  $id00006.A = a
  $id00006.B = $id00004.Y
  $id00006.Y = y
!end_macro sat

```

Figure: CircSat

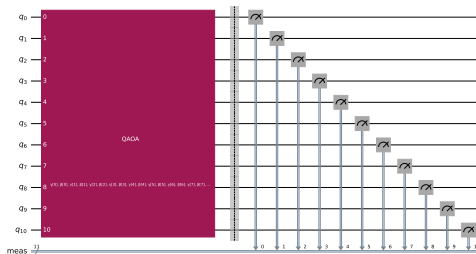
$$\begin{pmatrix}
 0.5 & 0.5 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 \\
 0 & 0.5 & -1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 1 & 0 & 0 & -2 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & -0.5 & 0.5 & -1 & -2 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & -0.5 & -1 & 0 & -2 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{pmatrix}$$


Figure: $\mathcal{H}_f$ and QAOA Quantum Circuit

Esempio

Risultato 1

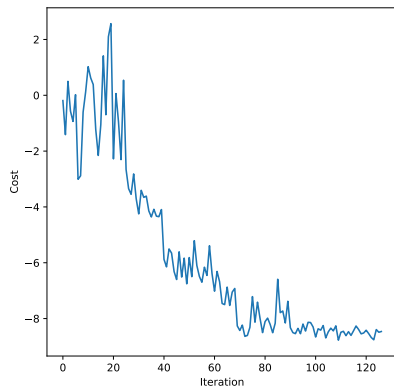


Figure: Parameter optimization

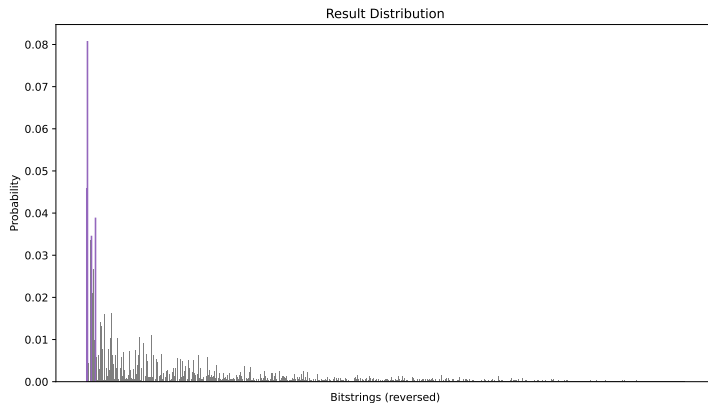


Figure: Sample results

Esempio

Risultato 2

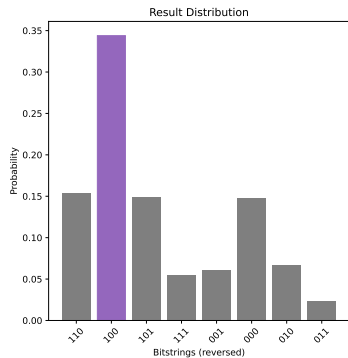


Figure: 3 “interesting” bit

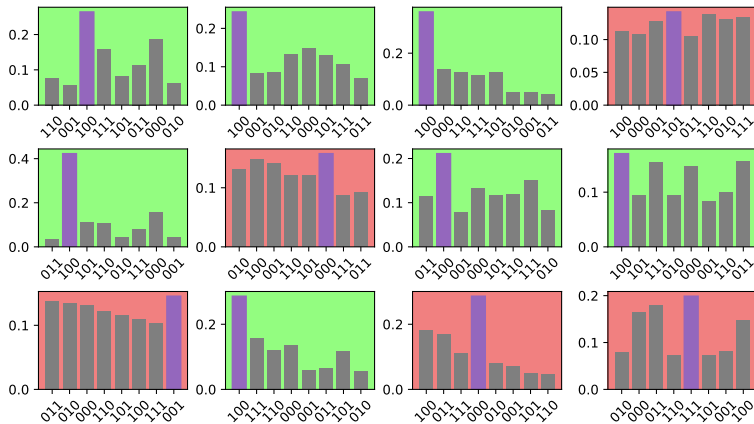


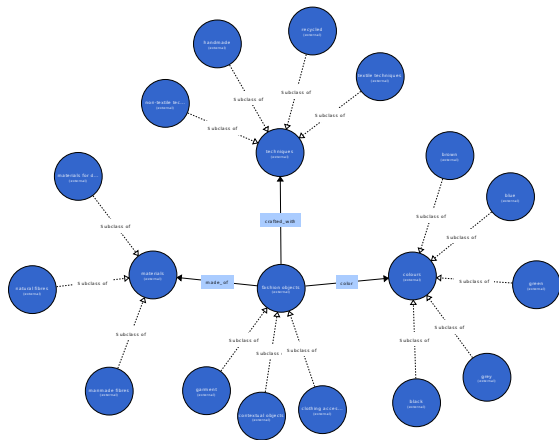
Figure: Multipe experiment runs

Cos'è un'ontologia

- Insieme di fatti
- Riguardanti un dominio di interesse
- Prevengono interpretazioni sbagliate
- Assicurano la cooperazione tra software

Come si definisce un'ontologia

- Ontology Web Language (OWL)
- Diversi “flavours”
 - OWL Full
 - OWL DL
- Classi Individui e Relazioni



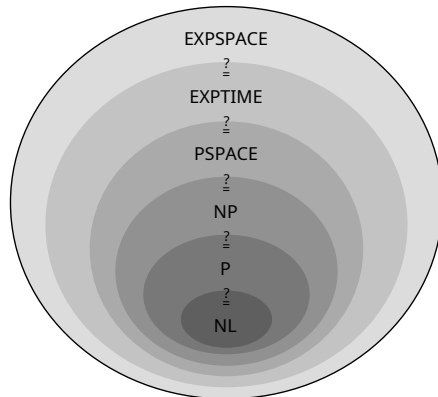
Inferenze in OWL

Complessità

- Variazione sintattica di *SROIQ*
- *ALC* è una restrizione di *SROIQ*
- *ALC* è PSpace-hard

Reasoner

- Pellet
- Fact++
- HermiT
- ...



Limitazioni del simulatore

```
cousins(P, Q):-  
    gran_parent_of(X, P),  
    gran_parent_of(X, Q),  
    parent_of(Y, P),  
    parent_of(Z, Q),  
    Z \= Y.
```

Figure: Cousins rule

Interrogazione

```
QA-Prolog --verbose --qasm-args="--solver=sim_anneal  
--postproc=opt" --query='cousins(P1, P2).' family.pl
```

```
QA-Prolog: No solutions were found
```

Figure: Query Result